

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from the Industrial Revolution to 2024

Scope: 1750 – 2024 | Sources: Global Carbon Budget 2025, NASA NOAA Mauna Loa Observatory

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

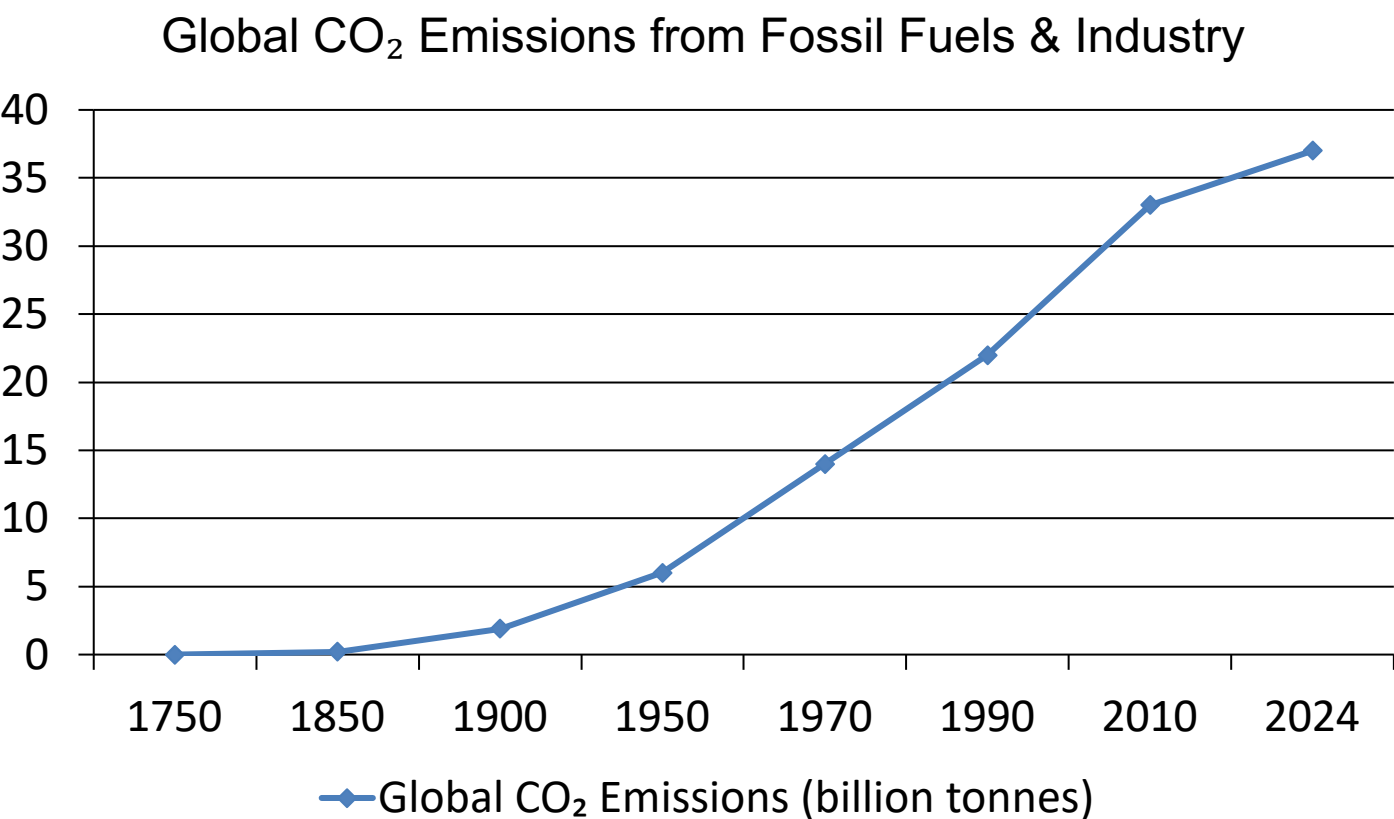
429 ppm

Latest measurement from NOAA Mauna Loa Observatory
(February 2026)

- Pre-industrial baseline: ~280 ppm
- Current level is >1.5× the natural increase observed at the end of the last ice age (20,000 years ago)
- That ancient change took thousands of years; the modern rise has occurred in roughly two centuries
- Ground-based and satellite measurements confirm sharp acceleration, linked primarily to burning coal, oil, and natural gas

Timeline: 280 ppm (pre-industrial) → 315 ppm (1958) → 429 ppm (2026)

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



1950: ~6 billion tonnes

Post-war industrial acceleration begins

1990: ~22 billion tonnes

Nearly quadrupled since 1950

2024: >35 billion tonnes

Growth slowed but not yet peaked

Reference: figure_2.jpg — Annual CO₂ emissions trajectory (Global Carbon Budget 2025)

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Annual CO ₂ (billion t)	Share of Global	Trend (recent)
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Key Insight

China's steep rise post-2000 has made it the world's largest emitter by a significant margin.

The US peaked near 6 billion tonnes around 2005 and has since declined to roughly 5 billion tonnes.

India has risen rapidly to approximately 3 billion tonnes and continues to grow.

Reference: figure_3.jpg — Annual CO₂ emissions by country (Global Carbon Budget 2025)

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900: Europe + US > 90%



Europe & US ~92% | Rest of World ~8%

1950: Europe + US Still > 85%



Europe & US ~87% | Rest of World ~13%

2024: Asia ~50%; US + Europe < 33%



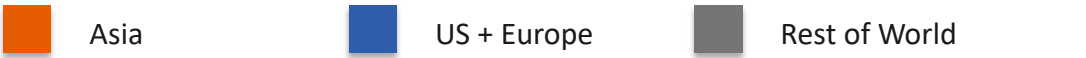
Asia ~50% | US + Europe ~32% | Rest ~18%

Regional Share Shift Visual

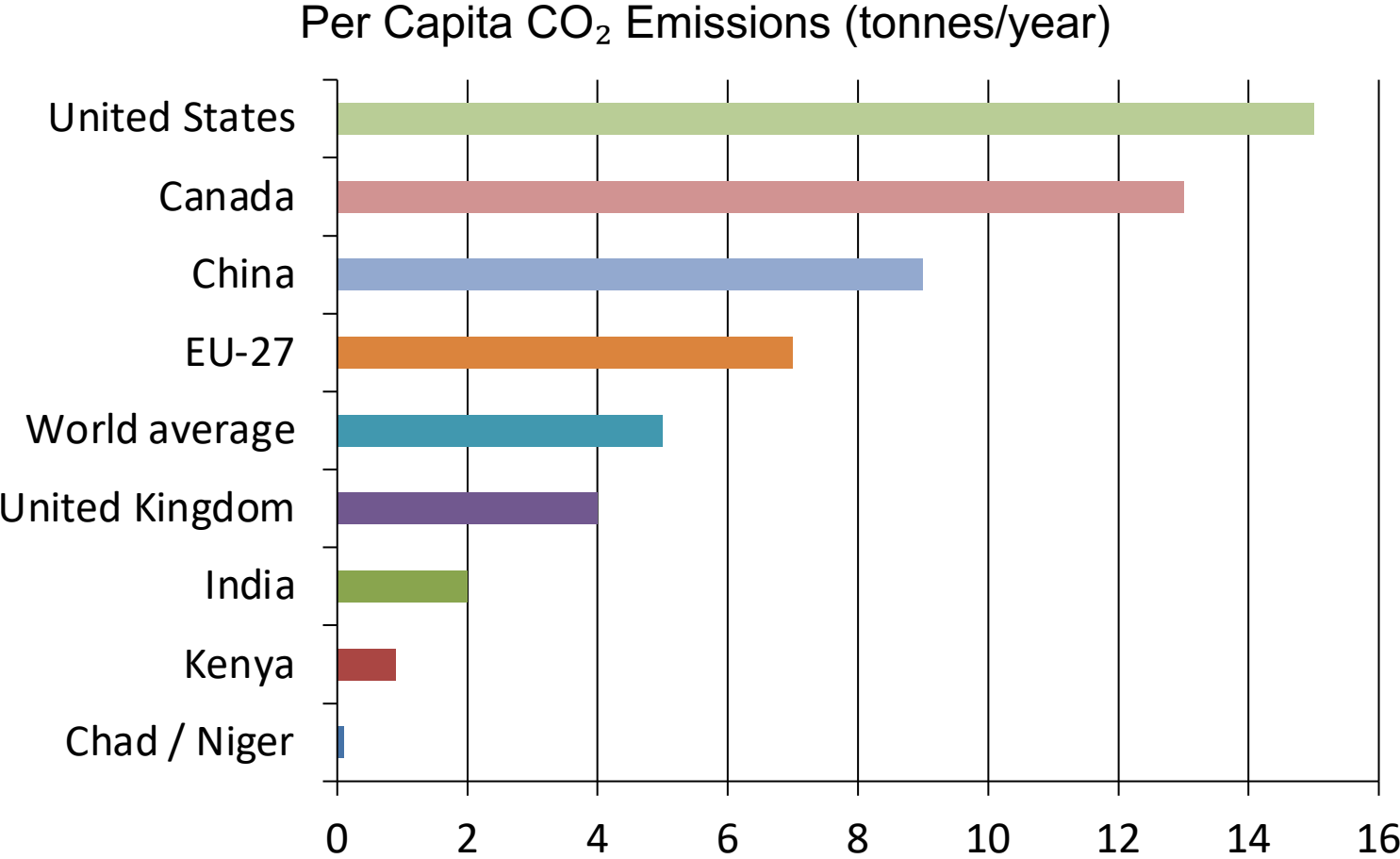
Stacked area chart showing the dramatic rebalancing of global emissions:

- 1900–1950: Industrialized West dominated (>85–90%)
- 1970–2000: Asia's share rises with manufacturing and urbanization
- 2024: Asia produces roughly half of all emissions, while the US and Europe combined fall below one-third

Africa and South America each contribute only 3–4% of global emissions — comparable to international aviation and shipping combined.



Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150× Disparity

The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.

Among large, populous countries, the US, Australia, and Canada have per capita emissions roughly three times the global average.

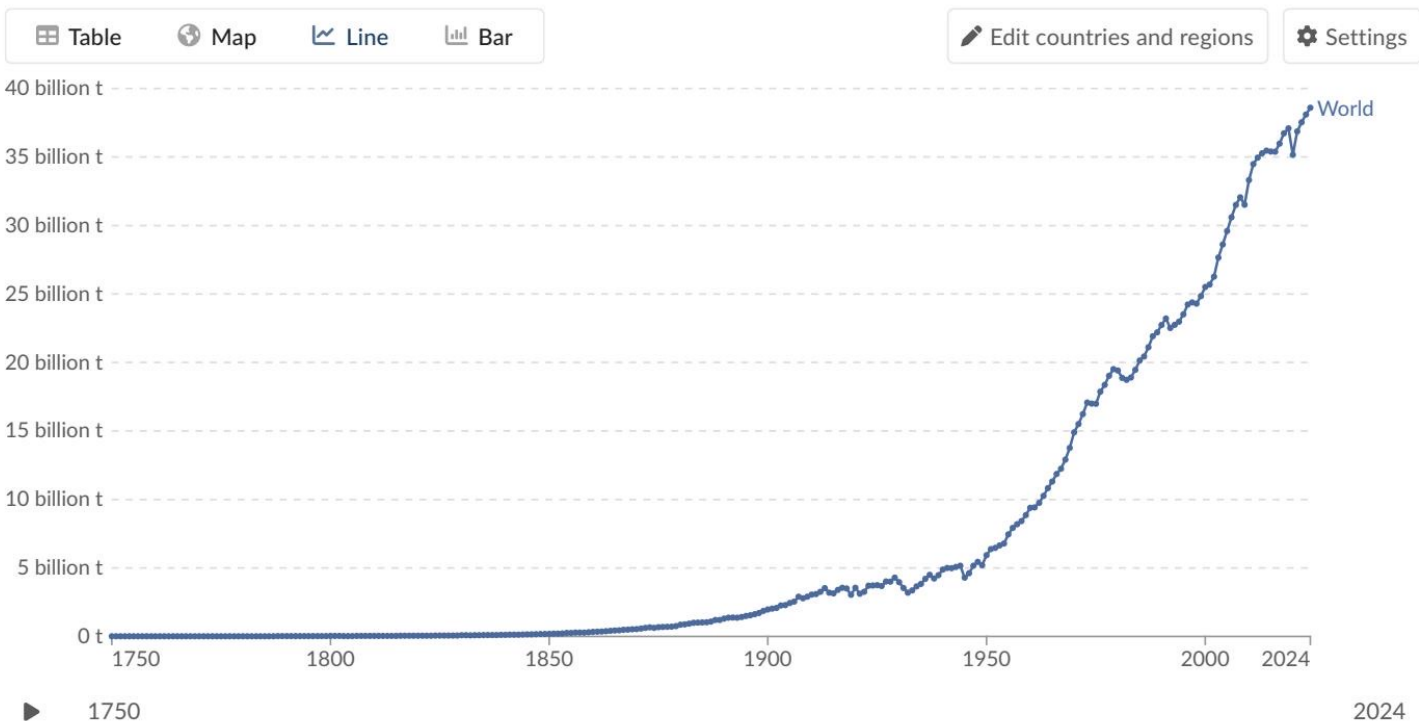
Reference: figure_1.jpg — Per capita CO₂ emissions over time (Our World in Data)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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2024 Annual % Change in CO₂ Emissions

- Blue (negative): US and much of Europe recorded declines of roughly –1% to –5%
- Orange / Red (positive): Russia, parts of Southeast Asia, and several African nations saw increases of +2% to +10% or more
- Deep red (>+10%): Some countries in South America experienced very large percentage increases
- Mixed picture: Not all regions move in the same direction — growth and decline coexist globally

Reference: figure_4.jpg — Annual percentage change in CO₂ emissions, 2024 (Global Carbon Budget 2025)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their energy sources.

Country	Nuclear & Renewables Share	Fossil Fuel Share	Per Capita CO ₂ (t)	Key Driver
France	High (nuclear dominant)	Low	~4 t	Nuclear power base
United Kingdom	High (renewables + nuclear)	Low	~4 t	Offshore wind, nuclear legacy
Germany	Moderate (renewables growing)	High (coal, gas)	Higher than FR/UK	Lignite & coal dependency
Netherlands	Moderate	High (gas, coal)	Higher than FR/UK	Natural gas & industrial demand

Policy implication: Energy mix and policy choices — not just prosperity — determine per capita outcomes. Nuclear and renewable transitions can decouple economic growth from emissions.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global emissions exceed 35 Bt and have not yet peaked

Despite slowing growth rates, annual CO₂ from fossil fuels and industry continues to rise.

2. China alone emits >25% of the global total

At ~12.5 billion tonnes, China is more than double the US (~5 Bt); India (~3 Bt) is third and rising fast.

3. Per capita inequalities remain extreme — 150× gap

The average American or Australian emits in under two days what the average person in Chad or Niger emits in a year.

4. US and Europe are declining, but historical cumulative responsibility remains large

Europe and the US accounted for >90% of emissions in 1900; today Asia produces ~50% while their combined share is <33%.

5. Policy and energy mix choices — not just prosperity — determine per capita outcomes

France and the UK show that high living standards with low per capita CO₂ are achievable through nuclear and renewable energy.